

Lawrence Degoma

lawrencedegoma02@gmail.com | Glendale, CA | (818) 245-0308 | www.linkedin.com/in/lawrencedegoma | <https://github.com/lawrenceDegoma>

EDUCATION

California State University of Long Beach – College of Engineering
Bachelor of Computer Science

Expected Dec 2026

Pasadena City College

June 2024

A.A. Engineering and Technology | A.S. Mathematics | A.S. Natural Science

SKILLS

Languages: Python, Swift, C++, JavaScript, TypeScript, C

Frameworks and Libraries: OpenCV, MediaPipe, React, Node.js, Tailwind, Firebase

Databases: MongoDB, DynamoDB, PostgreSQL

Operating Systems: MacOS, Linux

Domains: Human-Machine Interfaces (HMI), Real-Time Interaction Systems, Computer Vision, Platform Engineering

PROJECTS

HandtrackerHUD | Python, OpenCV, MediaPipe

<https://github.com/lawrenceDegoma/HandTrackerHUD>

- Designed a real-time human-machine interface translating hand gestures and voice input into control signals
- Implemented landmark tracking, gesture classification, and latency-aware interaction pipelines
- Integrated Spotify Web API and macOS window capture to enable hands-free system control
- Developed a custom HUD overlay with perspective transforms and dynamically rendered UI components
- Tuned interaction feedback loops to prioritize responsiveness, ergonomics, and intuitive control

MacMultipeer | Swift, MultipeerConnectivity

<https://github.com/lawrenceDegoma/MacMultipeer>

- Built a cross-platform screen sharing and control system across macOS and iOS using MultipeerConnectivity
- Implemented system-level permission handling with ScreenCaptureKit, TCC workflows, and sandboxed app configs
- Engineered error handling and fallback logic supporting modern ScreenCaptureKit and legacy CMIO capture paths
- Configured secure Xcode build pipelines including entitlements and code signing for App Sandbox compliance
- Built a permission state caching system using UserDefaults to reduce repeated system prompts to improve UX

Bartending Ordering System | React, Firebase

<https://justins-julep.vercel.app/>

- Engineered a real time ordering system to streamline operations by queuing customer drinks to a receiver interface
- Designed a consistent order queue system synchronized across devices using Firebase real-time updates
- Built a responsive React and Tailwind frontend optimized for fast input during live events
- Optimized backend logic for speed, reliability, and scalability under bursty traffic conditions
- Deployed the system for live use at private events, improving bartender workflow efficiency

Cafe Index Website | HTML, CSS, Js

<https://lawrencedegoma.github.io/sea-Round2-Project/>

- Built a responsive, framework-free catalog website for cafe recommendations in the greater LA/OC areas
- Implemented dynamic user interactions using Javascript to provide a seamless browsing experience
- Designed with creative inspiration from the Pokédex in Pokemon games, showcasing a unique gamified user interface
- Focused on clean, maintainable code and user-centric design principles.

HTML/CSS Parser | C++

https://github.com/lawrenceDegoma/HTML_Parser

- Developed a custom HTML/CSS parser and renderer in C++ using SFML
- Implemented a tree-based document model for efficient parsing and layout processing
- Built a full parsing-to-rendering pipeline translating HTML/CSS into graphical output
- Focused on performance optimization and modular design to support extensibility

PROFESSIONAL EXPERIENCE

Ishkaster Media

Full Stack Engineer Intern

Aug. 2024 – Jan. 2025

- Developed a dynamic marketplace, building both frontend and backend systems
- Utilized APIs from Amazon and Best Buy to populate the marketplace with real-time data
- Designed a user friendly UI using Shopify frontend, creating a smooth user experience
- Built backend systems in Node.js to manage API integrations
- Implemented data validation and security measures to ensure site reliability

Starbucks

Barista

Sep. 2021 – Present

- Prepared a variety of beverages from a menu of 60+ items, ensuring consistent quality and customer satisfaction
- Fostered a positive and welcoming atmosphere for customers and team members, enhancing overall morale
- Led and coordinated a team of baristas during high-traffic periods, optimizing workflow and minimizing delays
- Mentored new hires and provided ongoing support to veteran staff, contributing to a strong, cohesive team
- Cultivated a community-centric environment, positioning the store as a “third place” for customers

Individual Work

Private Tutor

Jul. 2022 – Sep. 2023

- Designed and implemented customized lesson plans with a focus on mathematics, tailored to individual learning styles
- Applied a philosophical approach to teaching, making complex concepts relatable and fostering deeper student engagement
- Encouraged a growth mindset, helping students build confidence and improve problem solving abilities
- Delivered measurable improvements in student performance and retention, with a proven track record of success
- Notably, several students received a perfect score of a 5 on the 2024 AP Calc AB exam, demonstrating effective tutoring